Is Form Real? Modeling Cricket Batting Performance with Markov Chains

DS5020 Final Project — Iteration 2

Aditya Vishwanath Adithya Anand Manavkrishna Manoj

Fall 2026 · Dr. Fatema Nafa

Project leader, Iteration 2: Adithya Anand

Contents

Iteration 2 — The Math Foundation **1**

1 Project overview **1**

2 Notation **2**

3 Definitions **2**

4 Connecting the data to the math **3**

5 Assumptions **4**

6 Iteration 2 — Math Questions **4**

6.1 Q1. The Markov property and the transition matrix 4

6.2 Q2. The estimate $\hat{P}_{ij} = n_{ij}/n_i$ 5

6.3 Q3. Chapman–Kolmogorov: $P^{(n)} = P^n$ 6

6.4 Q4. The steady state as a linear system 6

6.5 Q5. Absorbing chains: $N = (I - Q)^{-1} = \sum_k Q^k$, $t = N\mathbf{1}$, $B = NR$ 7

7 The algorithm as math **8**

8 Data description **8**

9 Simulation Check **10**

Iteration 2 — The Math Foundation

1 Project overview

Cricket is a bat-and-ball team sport. A *batter* tries to score *runs*; his turn at bat is an *innings*, which ends when he is *out*. Before each game a team picks 11 players; a player who played the previous game but is not picked is *dropped*. We use the Indian Premier League (IPL), a professional league whose games use the short T20 format. A batter who has scored well recently is said to be "in form", cricket's version of the "hot hand".

We model a batter's results as a Markov chain at two scales. Both chains use the same mathematics (Sections 3–7); only the meaning of a state changes.

Chain I (innings level, our main question). One step is one innings of a top-order batter. The transient states are **F** (Fail, 0–9 runs), **S** (Start, 10–29) and **B** (Big, 30+); the absorbing state **D** means he is not in his team's next match. Question: does the state of today's innings change the distribution of the next one?

Chain II (ball level, as described in the course handout). One step is one ball faced. The transient states are the batter's runs off the ball, $0, 1, 2, 3, 4, 6$; the absorbing state is **Out**. Question: does the outcome of one ball change the distribution of the next, and how many balls does a batter face before he is out?

2 Notation

Table 1: Notation. Chain I has $k = 3$ transient states, Chain II has $k = 6$; both have $r = 1$ absorbing state.

Symbol	Meaning	Dimension / type
$\mathcal{S} = \{1, \dots, k\}$	transient states (Chain I: F, S, B; Chain II: 0, 1, 2, 3, 4, 6)	$k \in \{3, 6\}$
a	the absorbing state (D or Out), numbered $k + 1$	
X_t	state at step t (innings t of a batter, or ball t of an innings)	$X_t \in \mathcal{S} \cup \{a\}$
P	transition matrix among transient states, $P_{ij} = \Pr(X_{t+1} = j \mid X_t = i)$	$\mathbb{R}^{k \times k}$
$\bar{P}$	full transition matrix with the absorbing state, in canonical form	$\mathbb{R}^{(k+1)\times(k+1)}$
$Q,\ R$	blocks of $\bar{P}$: transient→transient, transient→absorbing	$\mathbb{R}^{k\times k},\ \mathbb{R}^{k\times 1}$
n_{ij}	number of observed transitions from i to j	$\mathbb{N}$
$n_i = \sum_j n_{ij}$	number of observed transitions out of i	$\mathbb{N}$
n	count matrix (n_{ij})	$\mathbb{N}^{k\times(k+1)}$
$\hat{P},\ \hat{Q},\ \hat{R}$	estimates from counts	as P, Q, R
$P_{ij}^{(n)}$	n-step probability $\Pr(X_{t+n} = j \mid X_t = i)$	$\mathbb{R}$
π	steady state (row vector)	$\mathbb{R}^{1\times k}$
$\mathbf{1},\ e_k$	vector of ones; last standard basis vector	$\mathbb{R}^k$
A	steady-state system matrix (Section 6.4)	$\mathbb{R}^{k\times k}$
N	fundamental matrix $(I - Q)^{-1}$	$\mathbb{R}^{k\times k}$
τ_i	number of steps until absorption, starting in i (random)	$\{1, 2, \dots\}$
t	expected absorption times, $t_i = \mathbb{E}[\tau_i]$	$\mathbb{R}^k$
B	absorption probabilities NR	$\mathbb{R}^{k\times 1}$
$\ell(P)$	log-likelihood of the observed sequences	$\mathbb{R}$
λ_i	Lagrange multiplier for row i	$\mathbb{R}$
se_{ij}	standard error of $\hat{P}_{ij}$	$\mathbb{R}$
X^2, df	chi-squared statistic and its degrees of freedom	$\mathbb{R},\ \mathbb{N}$
T	length of a simulated path	$\mathbb{N}$

3 Definitions

Definition 1 (Conditional probability)**.** For events E, F with $\Pr(F) > 0$, $\Pr(E \mid F) = \Pr(E \cap F)/\Pr(F)$. For fixed F, $\Pr(\cdot \mid F)$ is a probability measure: it is non-negative, $\Pr(\Omega \mid F) = 1$, and it is additive over disjoint events.

Definition 2 (Markov chain, time-homogeneous)**.** A sequence $X_0, X_1, \dots$ with values in a finite set is a *Markov chain* if for all t and all states $\Pr(X_{t+1} = j \mid X_t = i, X_{t-1} = x_{t-1}, \dots, X_0 = x_0) = \Pr(X_{t+1} = j \mid X_t = i)$. It is *time-homogeneous* if this probability does not depend on t.

Definition 3 (Stochastic matrix, probability vector)**.** A square matrix with non-negative entries whose rows each sum to 1 is *(row-)stochastic*. A row vector with non-negative entries summing to 1 is a *probability vector*.

Definition 4 (n-step transition probability)**.** $P_{ij}^{(n)} = \Pr(X_{t+n} = j \mid X_t = i)$; for a time-homogeneous chain it does not depend on t.

Definition 5 (Steady state)**.** A probability vector π with $\pi P = \pi$ is a *steady state* (stationary distribution). Equivalently, $\pi^\top$ is an eigenvector of $P^\top$ for the eigenvalue 1 (a *left* eigenvector of P).

Definition 6 (Eigenvalue, eigenvector, rank, null space)**.** λ is an eigenvalue of M with eigenvector $v \neq 0$ if $Mv = \lambda v$. The null space is $N(M) = \{v : Mv = 0\}$; $\operatorname{rank} M + \dim N(M) =$ number of columns (rank–nullity theorem). M (square) is invertible iff $N(M) = \{0\}$ iff $\det M \neq 0$.

Definition 7 (Irreducible, aperiodic, regular)**.** A chain is *irreducible* if every state can be reached from every other state, and *aperiodic* if returns to a state are not restricted to multiples of some $d > 1$. A stochastic matrix is *regular* if some power P^m has all entries positive; a regular chain is irreducible and aperiodic.

Definition 8 (Absorbing state and absorbing chain; canonical form)**.** State a is *absorbing* if $\bar{P}_{aa} = 1$. A chain is *absorbing* if it has at least one absorbing state and an absorbing state can be reached from every state. Ordering the transient states first,

$$\bar{P} = \begin{bmatrix} Q & R \\ 0 & I \end{bmatrix}.$$

Definition 9 (Fundamental matrix, absorption times and probabilities)**.** $N = (I - Q)^{-1}$; $t = N\mathbf{1}$; $B = NR$ (meaning derived in Proposition 5).

Definition 10 (Likelihood and maximum-likelihood estimate)**.** The likelihood $L(\theta)$ is the probability of the observed data as a function of the parameters θ; the MLE maximizes L (equivalently $\ell = \log L$, since log is strictly increasing).

Definition 11 (Standard error of a proportion)**.** If $\hat{p}$ is the fraction of n independent trials with success probability p, then $\mathrm{Var}(\hat{p}) = p(1-p)/n$ and $\mathrm{se}(\hat{p}) = \sqrt{\hat{p}(1-\hat{p})/n}$; by the central limit theorem $\hat{p} \pm 1.96\,\mathrm{se}$ is an approximate 95% interval.

4 Connecting the data to the math

The data are Cricsheet's ball-by-ball records of all 1,243 IPL matches (2008-04-18 to 2026-05-31).

- **Chain I.** One row of the innings table is one innings of one of the 74 pool batters (top-order batters with at least 40 innings). Its *state* is F, S or B from his runs. One *step* goes from this innings to his team's next match in the same season. One *count* n_{ij} is the number of times an innings in state i was followed, in the team's next match, by an innings in state j ($j \in \{F, S, B\}$) or by his omission from the team ($j = D$). One row of $n \in \mathbb{N}^{3\times 4}$ contains all transitions out of one state; one column contains all transitions into one state.

- **Chain II.** One row of the ball table is one ball faced by a pool batter. Its state is his runs off that ball (0, 1, 2, 3, 4 or 6), or Out if he was dismissed on it. One *sequence* is one innings; n_{ij} counts consecutive pairs of balls in the same innings with outcomes i then j, and $n \in \mathbb{N}^{6\times 7}$.

- In both chains a sequence can also end without absorption (*censoring*): the season ends, or the batter is not out when his team's innings ends. Censored sequences contribute their transitions but no transition into a.

5 Assumptions

Table 2: Assumptions, what goes wrong if they fail, and how we check them in Iteration 3.

Assumption	What goes wrong in the math if it fails	Check in Iteration 3
A1 Markov property	The likelihood no longer factors as $\prod P_{ij}^{n_{ij}}$, so $\hat{P}$ is not the MLE; Chapman–Kolmogorov fails, so $\hat{P}^n$ is not the n-step probability.	Compare $\Pr(X_{t+1} \mid X_t, X_{t-1})$ across X_{t-1}; chi-squared test.
A2 Time-homogeneous	P depends on t; one $\hat{P}$ averages different matrices and $\hat{P}^n$, π and N describe no actual period.	Estimate P for 2008–2016 and 2017–2026 separately; homogeneity test.
A3 Regular (irreducible, aperiodic)	π may not be unique and P^n may not converge, so the steady-state system is singular even with $\sum \pi_i = 1$.	All entries of $\hat{P}^2$ positive.
A4 Enough data in each row	If $n_i = 0$ the row of $\hat{P}$ is undefined; if n_i is small the normal-approximation intervals are poor.	Report $\min_i n_i$.
A5 Absorption reachable from every transient state	$Q^n \not\to 0$, $I - Q$ can be singular, and t is infinite.	Every row sum of $\hat{Q}$ below 1.
A6 Same P for every batter (pooling)	A mixture of chains is not a Markov chain; pooling good and bad batters can create apparent dependence.	Chi-squared test of batter $\times$ next state, per row.
A7 Independent sequences, non-informative censoring	The likelihood is no longer a product over sequences; if censoring depends on the future, counts are biased.	Argued from the data design (separate innings, season ends fixed in advance).
A8 "Not in the next match" means dropped	D also absorbs injuries and rest, so t under-states time to a selection decision.	Not checkable from scorecards; stated as a limitation.

6 Iteration 2 — Math Questions

The five questions below are our derivations (the handout requires at least three). Each result is stated as a proposition and proved; the rule used at each step is shown in blue.

6.1 Q1. The Markov property and the transition matrix

Proposition 1. *For a time-homogeneous Markov chain define $P_{ij} = \Pr(X_{t+1} = j \mid X_t = i)$. Then P is row-stochastic.*

Proof. The Markov property says the next state depends on the past only through the present: $\Pr(X_{t+1} = j \mid X_t = i, X_{t-1}, \ldots, X_0) = \Pr(X_{t+1} = j \mid X_t = i)$, and time-homogeneity says this

number is the same for every t, so P_{ij} is well defined. Then

$$P_{ij} = \Pr(X_{t+1} = j \mid X_t = i) \ \geq 0$$

[conditional probability is a probability, axiom of non-negativity]

$$\begin{aligned}\sum_j P_{ij} &= \sum_j \Pr(X_{t+1} = j \mid X_t = i) \\ &= \Pr\left(\bigcup_j \{X_{t+1} = j\} \;\middle|\; X_t = i\right) \\ &= \Pr(\Omega \mid X_t = i) = 1\end{aligned}$$

[additivity: the events $\{X_{t+1} = j\}$ are disjoint]

[the events cover every possible next state]

so each row of P is a probability distribution. □

Owner: Aditya Vishwanath *Checked by: Manavkrishna Manoj*

6.2 Q2. The estimate $\hat{P}_{ij} = n_{ij}/n_i$

Proposition 2. *Given observed sequences from a Markov chain with $n_i > 0$ for every i, the maximum-likelihood estimate is $\hat{P}_{ij} = n_{ij}/n_i$, and each row of $\hat{P}$ sums to 1.*

Proof. For one sequence $x_{0:T} = (x_0, \ldots, x_T)$:

$$\begin{aligned}\Pr(X_{0:T} = x_{0:T}) &= \Pr(X_0 = x_0) \prod_{t=0}^{T-1} \Pr(X_{t+1} = x_{t+1} \mid X_{0:t} = x_{0:t}) \\ &= \Pr(X_0 = x_0) \prod_{t=0}^{T-1} P_{x_t x_{t+1}} \\ &= \Pr(X_0 = x_0) \prod_{i,j} P_{ij}^{n_{ij}}\end{aligned}$$

[multiplication (chain) rule]

[Markov property, time-homogeneity]

[group equal factors; n_{ij} counts them]

For several sequences the probabilities multiply (A7, independence), and the counts add, so the same form holds with the total counts. Conditioning on the first state of each sequence, the log-likelihood is

$$\ell(P) = \sum_{i,j} n_{ij} \log P_{ij}$$

[log of a product is a sum of logs]

to be maximized subject to $\sum_j P_{ij} = 1$ for each i. The Lagrangian is $\mathcal{L} = \sum_{i,j} n_{ij} \log P_{ij} - \sum_i \lambda_i \big(\sum_j P_{ij} - 1\big)$, and

$$\frac{\partial \mathcal{L}}{\partial P_{ij}} = \frac{n_{ij}}{P_{ij}} - \lambda_i = 0 \implies P_{ij} = \frac{n_{ij}}{\lambda_i}$$

[Lagrange condition, $\frac{d}{dp} \log p = 1/p$]

$$\sum_j P_{ij} = 1 \implies \frac{1}{\lambda_i} \sum_j n_{ij} = 1 \implies \lambda_i = n_i$$

[substitute into the constraint]

$$\implies \hat{P}_{ij} = \frac{n_{ij}}{n_i}.$$

ℓ is a sum of concave functions (log is concave) and the constraints are linear, so this stationary point is the global maximum [concave function on a convex set]. If $n_{ij} = 0$ the term is absent and $\hat{P}_{ij} = 0$ maximizes ℓ on the boundary. Finally $\sum_j \hat{P}_{ij} = \frac{1}{n_i} \sum_j n_{ij} = \frac{n_i}{n_i} = 1$, so every row of $\hat{P}$ is a probability distribution; this needs $n_i > 0$ (A4). □

Owner: Adithya Anand *Checked by: Aditya Vishwanath*

6.3 Q3. Chapman–Kolmogorov: $P^{(n)} = P^n$

Proposition 3. *For a time-homogeneous Markov chain, $P_{ij}^{(n)} = (P^n)_{ij}$ for all $n \geq 1$.*

Proof. By induction on n. *Base case* $n = 1$: $P_{ij}^{(1)} = \Pr(X_{t+1} = j \mid X_t = i) = P_{ij}$ [definition]. *Inductive step*: assume $P^{(n)} = P^n$. Then

$$
\begin{aligned}
P_{ij}^{(n+1)} &= \Pr(X_{t+n+1} = j \mid X_t = i) \\
&= \sum_l \Pr(X_{t+n+1} = j,\ X_{t+n} = l \mid X_t = i) && \text{[law of total probability over } X_{t+n}\text{]} \\
&= \sum_l \Pr(X_{t+n+1} = j \mid X_{t+n} = l,\ X_t = i)\, \Pr(X_{t+n} = l \mid X_t = i) && \text{[multiplication rule]} \\
&= \sum_l P_{lj}\, P_{il}^{(n)} && \text{[Markov property, time-homogeneity]} \\
&= \sum_l (P^n)_{il}\, P_{lj} && \text{[induction hypothesis]} \\
&= (P^n P)_{ij} = (P^{n+1})_{ij} && \text{[definition of the matrix product]}
\end{aligned}
$$

□

Remark. The same argument gives $P^{(m+n)} = P^{(m)} P^{(n)}$, the Chapman–Kolmogorov equation. Each row of P^n is a probability vector, since a product of stochastic matrices is stochastic ($P^n \mathbf{1} = P^{n-1}(P\mathbf{1}) = \mathbf{1}$).

Owner: Manavkrishna Manoj *Checked by: Adithya Anand*

6.4 Q4. The steady state as a linear system

Proposition 4. *(a) $\pi P = \pi$ is equivalent to $(P^\top - I)\pi^\top = 0$, and $P^\top - I$ is singular, so this system alone does not determine π. (b) If P is regular, replacing the last equation by $\mathbf{1}^\top \pi^\top = 1$ gives a nonsingular matrix A, and $A\pi^\top = e_k$ has exactly one solution, the steady state.*

Proof. (a) Transposing, $\pi P = \pi \iff P^\top \pi^\top = \pi^\top \iff (P^\top - I)\pi^\top = 0$ [$(XY)^\top = Y^\top X^\top$]. Because every row of P sums to 1 (Proposition 1), $P\mathbf{1} = \mathbf{1}$, so $(P - I)\mathbf{1} = 0$ with $\mathbf{1} \neq 0$: $P - I$ is singular, and $\det(P^\top - I) = \det\big((P - I)^\top\big) = \det(P - I) = 0$ [$\det M^\top = \det M$]. Equivalently, the rows of $P^\top - I$ add to zero, $\mathbf{1}^\top (P^\top - I) = \big((P - I)\mathbf{1}\big)^\top = 0$, so any one equation is the negative sum of the others and is redundant. If $\pi^\top$ solves the system so does $c\,\pi^\top$ for every c, so the solution is not unique; this is also why 1 is an eigenvalue of P and hence of $P^\top$ [P and $P^\top$ have the same characteristic polynomial].

(b) For a regular stochastic matrix the eigenvalue 1 is simple and has a positive eigenvector [Perron–Frobenius theorem], so $\dim N(P^\top - I) = 1$ and $N(P^\top - I) = \{c\,\pi^\top\}$ with π a probability vector. Let A be $P^\top - I$ with its last row replaced by $\mathbf{1}^\top$, and suppose $Av = 0$. The first $k-1$ entries of $(P^\top - I)v$ are zero, and its last entry is minus their sum (rows add to zero), so $(P^\top - I)v = 0$ and $v = c\,\pi^\top$. The last row of $Av = 0$ gives $0 = \mathbf{1}^\top v = c\,\mathbf{1}^\top \pi^\top = c$, so $v = 0$. Hence $N(A) = \{0\}$ and A is invertible [invertible iff trivial null space], so $A\pi^\top = e_k$ has the unique solution $\pi^\top = A^{-1}e_k$, which satisfies both $\pi P = \pi$ and $\sum_i \pi_i = 1$. □

Owner: Aditya Vishwanath *Checked by: Adithya Anand*

6.5 Q5. Absorbing chains: $N = (I-Q)^{-1} = \sum_k Q^k$, $t = N\mathbf{1}$, $B = NR$

Proposition 5. *Let $\bar{P}$ be in canonical form and suppose every row of Q sums to less than 1 (A5). Then (a) $\bar{P}^n = \begin{bmatrix} Q^n & (I+Q+\cdots+Q^{n-1})R \\ 0 & I \end{bmatrix}$; (b) $I-Q$ is invertible and $N = (I-Q)^{-1} = \sum_{m\geq 0} Q^m$; (c) N_{ij} is the expected number of visits to j starting from i, and $t = N\mathbf{1}$ gives $t_i = \mathbb{E}[\tau_i]$; (d) $B = NR$ gives the absorption probabilities, and with one absorbing state $B = \mathbf{1}$.*

Proof. (a) By induction: $n = 1$ is the canonical form. If it holds for n, block multiplication gives

$$\bar{P}^{n+1} = \bar{P}^n \bar{P} = \begin{bmatrix} Q^n Q & Q^n R + (I+\cdots+Q^{n-1})R \\ 0 & I \end{bmatrix} = \begin{bmatrix} Q^{n+1} & (I+\cdots+Q^n)R \\ 0 & I \end{bmatrix}$$

[block matrix product].

(b) Let $q = \max_i \sum_j Q_{ij} < 1$, which is $\|Q\|_\infty$ for a non-negative matrix. Then $\|Q^m\|_\infty \leq \|Q\|_\infty^m = q^m \to 0$ [the induced ∞-norm is submultiplicative], so $Q^m \to 0$. The partial sums satisfy

$$(I-Q)(I+Q+\cdots+Q^{m-1}) = I - Q^m \qquad \text{[telescoping sum]}$$

The entries of $\sum_m Q^m$ are non-negative and bounded by $\sum_m q^m = 1/(1-q)$, so the series converges [comparison with a geometric series]. Letting $m \to \infty$, $(I-Q)\sum_{m\geq 0} Q^m = I$, so $I-Q$ is invertible with inverse $N = \sum_{m\geq 0} Q^m$.

(c) For transient i, j, by (a) and Proposition 3, $\Pr(X_m = j \mid X_0 = i) = (\bar{P}^m)_{ij} = (Q^m)_{ij}$. Hence

$$\mathbb{E}\Big[\sum_{m\geq 0} \mathbf{1}\{X_m = j\} \;\Big|\; X_0 = i\Big] = \sum_{m\geq 0} \Pr(X_m = j \mid X_0 = i) \qquad \text{[linearity of expectation, } \mathbb{E}[\mathbf{1}_E] = \Pr(E)\text{]}$$

$$= \sum_{m\geq 0} (Q^m)_{ij} = N_{ij}. \qquad \text{[part (b)]}$$

The chain spends one step in a transient state at each of $X_0, \ldots, X_{\tau_i - 1}$, so τ_i is the total number of visits to transient states, and $t_i = \mathbb{E}[\tau_i] = \sum_j N_{ij} = (N\mathbf{1})_i$ [sum of the expected visits over j].

(d) The chain is absorbed in a for the first time at step $m+1$ exactly when $X_m = j$ is transient and $X_{m+1} = a$; these events are disjoint over (m, j). So

$$\Pr(\text{absorbed in } a \mid X_0 = i) = \sum_{m\geq 0}\sum_j (Q^m)_{ij} R_{ja} = (NR)_{ia}$$

[additivity over the disjoint events, multiplication rule, Markov property]. With one absorbing state, each row of $[Q\ R]$ sums to 1, so $R = \mathbf{1} - Q\mathbf{1} = (I-Q)\mathbf{1}$ and $B = N(I-Q)\mathbf{1} = \mathbf{1}$: absorption is certain. □

Remark. t also solves the first-step equation $t = \mathbf{1} + Qt$ (one step, then the expected time from where the chain moved), which is $(I - Q)t = \mathbf{1}$. In Chain I, t_i is the expected number of innings, counting the current one, before the batter is left out. In Chain II, t_i is the expected number of further balls he faces, counting the ball he is out on.

Owner: Adithya Anand *Checked by: Manavkrishna Manoj*

7 The algorithm as math

Input: sequences of states with L observed transitions in total; k transient states and one absorbing state. Chain I has $k = 3$, $L \approx 6{,}000$; Chain II has $k = 6$, $L \approx 140{,}000$.

(1) **Count.** $n_{ij} = \#\{t : X_t = i,\ X_{t+1} = j\}$ over all sequences, never across two sequences; $n \in \mathbb{N}^{k\times(k+1)}$. Cost: L additions.

(2) **Estimate** (Q2). $n_i = \sum_j n_{ij}$; $\hat{\hat{P}}_{ij} = n_{ij}/n_i$, and $\hat{P}$ from the $k \times k$ block of counts. Cost: $k(k+1)$ additions and divisions.

(3) **Uncertainty.** $\mathrm{se}_{ij} = \sqrt{\hat{P}_{ij}(1 - \hat{P}_{ij})/n_i}$. Cost: $O(k^2)$.

(4) **n-step probabilities** (Q3). $\hat{P}^n$ by repeated squaring, $\hat{P}^{2m} = (\hat{P}^m)^2$, $\hat{P}^{2m+1} = \hat{P}\,\hat{P}^{2m}$: at most $2\lceil \log_2 n \rceil$ products of $k \times k$ matrices, $2k^3$ flops each.

(5) **Steady state** (Q4). $A \in \mathbb{R}^{k\times k}$: rows $1, \ldots, k-1$ of $\hat{P}^\top - I$, last row $\mathbf{1}^\top$; right-hand side e_k. Gaussian elimination with partial pivoting on $[A \mid e_k]$ (about $\frac{2}{3}k^3$ flops), then back-substitution (k^2 flops) gives $\pi^\top$.

(6) **Canonical form.** $\hat{Q} = \hat{\hat{P}}_{\mathcal{S},\mathcal{S}} \in \mathbb{R}^{k\times k}$, $\hat{R} = \hat{\hat{P}}_{\mathcal{S},a} \in \mathbb{R}^{k\times 1}$.

(7) **Fundamental matrix** (Q5). Solve $(I - \hat{Q})N = I$ by Gaussian elimination on the $k \times 2k$ matrix $[I - \hat{Q} \mid I]$ and back-substitution for each of the k columns: about $\frac{8}{3}k^3$ flops.

(8) **Absorption.** $t = N\mathbf{1}$ (k^2 flops), $B = N\hat{R}$ ($2k^2$ flops), $\mathrm{Var}(\tau) = (2N - I)t - t \circ t$ ($2k^2$ flops).

(9) **Markov check.** Second-order counts n_{hij} (L additions) and a chi-squared homogeneity statistic for each current state i ($O(k^3)$).

With $k \le 6$ every matrix step costs under 10^3 flops; the cost is dominated by reading the data ($O(L)$).

8 Data description

Cricsheet (`https://cricsheet.org`) provides one JSON file per match (1,243 matches, 19 seasons, 2008-04-18 to 2026-05-31), released under the ODC-By licence. `src/load_data.py` builds an *appearances table* of 27,909 rows (one per player per team per match) and 11 columns (Table 3), covering 809 players. The only missing values are batting position, runs and balls for the 8,798 appearances in which the player did not bat; these are structural (nothing to record), not lost data.

Table 3: The appearances table built by `load_data.py` (one row per player per team per match).

Column	Type	Missing	Meaning
`match_id`	`str`	0	Cricsheet match identifier
`date`	`datetime64[us]`	0	match date
`season`	`int64`	0	IPL season (year of the match date)
`team`	`str`	0	franchise name
`player_id`	`str`	0	Cricsheet registry id (unique per person)
`player`	`str`	0	player name
`batted`	`bool`	0	did he bat in this match?
`position`	`float64`	8,798	batting position (order at the crease)
`runs`	`float64`	8,798	runs scored by the batter
`balls`	`float64`	8,798	balls faced (wides excluded)
`game_no`	`int64`	0	index of this match within the team's season

Parsing decisions: the season is taken from the match date (Cricsheet labels 2008, 2010 and 2020 as "2007/08", "2009/10" and "2020/21"); batting position is the order in which players first come to the crease; super-over innings are ignored; from 2023 a team's player list includes its substitute (the "Impact Player"), who counts as selected. The pool has 74 batters and 7,112 innings (2,016 F, 2,249 S, 2,847 B). In Chain I, 362 innings are followed by a drop, 453 are the team's last match of a season (censored) and 580 are followed by a match in which the batter played but had no top-4 innings (sequence break). For Chain II there are 7,112 innings with 143,951 balls faced: 6,378 end in a dismissal and 734 are censored. Wides (9,871 in the archive) are not balls faced and are skipped; no-balls (563 faced by pool batters) are balls faced and keep the batter's runs; the 74 five-run balls in the archive are counted with 4.

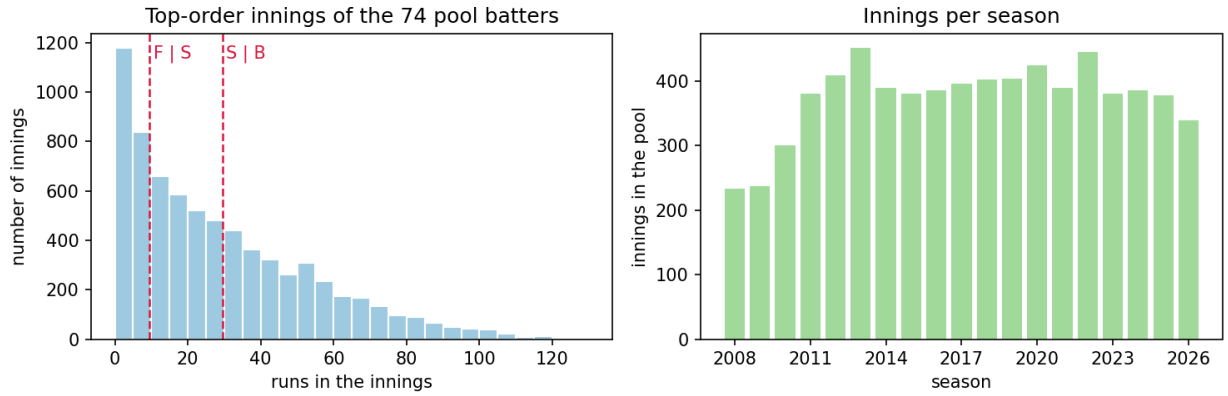


Figure 1: Left: runs per innings for the pool batters, with the F/S/B thresholds. Right: pool innings per season.

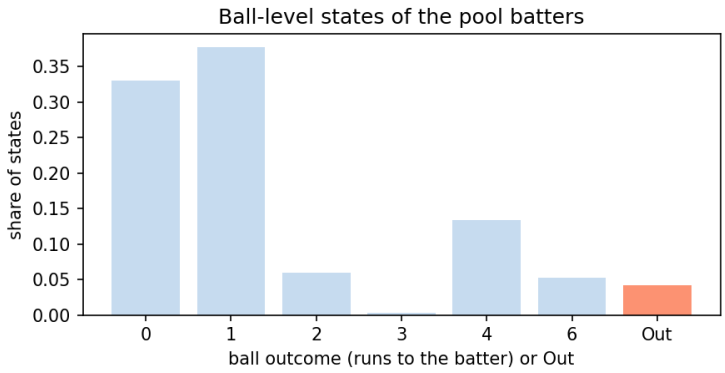


Figure 2: Share of each ball-level state for the pool batters.

9 Simulation Check

Model. A three-state Markov chain with the handout's transition matrix

$$P = \begin{bmatrix} 0.6 & 0.3 & 0.1 \\ 0.2 & 0.5 & 0.3 \\ 0.1 & 0.4 & 0.5 \end{bmatrix},$$

one path of length $T = 20{,}000$ from $X_0 = 0$, seed `np.random.default_rng(0)`. For Q5 we add an absorbing state reached from states $0, 1, 2$ with probabilities $(0.15, 0.05, 0.02)$ (row i of the full matrix is $(1 - p_i)P_{i\cdot}$ followed by p_i) and simulate 20,000 paths from a uniformly chosen start. For Q3 we simulate 20,000 paths of 3 steps from each state. Code: `python src/simulate.py`.

Table 4: Simulation check, Q2: transition matrix estimated from one path of length $T = 20{,}000$.

Entry	True	Estimate $\hat{P}_{ij}$	$\lvert\hat{P}_{ij} - P_{ij}\rvert/\text{se}$
P_{00}	0.60	0.594	0.94
P_{01}	0.30	0.304	0.66
P_{02}	0.10	0.102	0.53
P_{10}	0.20	0.197	0.71
P_{11}	0.50	0.497	0.55
P_{12}	0.30	0.306	1.22
P_{20}	0.10	0.106	1.69
P_{21}	0.40	0.399	0.20
P_{22}	0.50	0.495	0.82

Table 5: Simulation check, Q4: steady state.

	True π (from P)	From $\hat{P}$	Long-run frequency
π_0	0.2826	0.2805	0.2806
π_1	0.4130	0.4127	0.4127
π_2	0.3043	0.3068	0.3068

Table 6: Simulation check, Q3: 3-step transition probabilities.

(i, j)	$(P^3)_{ij}$	Fraction of 20,000 simulated paths
$(0, 0)$	0.3520	0.3514
$(0, 1)$	0.3940	0.3962
$(0, 2)$	0.2540	0.2523
$(1, 0)$	0.2680	0.2711
$(1, 1)$	0.4180	0.4129
$(1, 2)$	0.3140	0.3160
$(2, 0)$	0.2380	0.2359
$(2, 1)$	0.4240	0.4304
$(2, 2)$	0.3380	0.3337

Table 7: Simulation check, Q5: expected number of steps before absorption (20,000 simulated paths).

Start state	Pr(absorbed next)	$t = N\mathbf{1}$	Simulated mean (95% CI)	$B = NR$
0	0.15	12.87	12.98 ± 0.34	1.0000
1	0.05	15.40	15.46 ± 0.35	1.0000
2	0.02	16.27	16.13 ± 0.35	1.0000

- **Q1:** every row of $\hat{P}$ sums to 1 (largest deviation 0.000), as a stochastic matrix must.
- **Q2:** every $\hat{P}_{ij}$ is within 1.69 standard errors of the true P_{ij}, so the count ratio recovers P.
- **Q3:** $(P^3)_{ij}$ matches the fraction of simulated 3-step paths to within 0.006.
- **Q4:** $P^\top - I$ alone has rank 2 (singular); with the sum-to-one row the solution matches the long-run frequencies to within 8.2×10^{-5}.
- **Q5:** $t = N\mathbf{1}$ matches the simulated mean time to absorption to within 0.13 steps, and $B = NR = \mathbf{1}$.